Using Python in Music Sync

Objective: To ingest a music catalog into Python and execute algorithmic track matching against creative sync briefs using engineered metadata.

Method:

- Exported playlist catalog metadata from Apple Music into a raw CSV. Ingested the data into Python (using Pandas) for initial cleaning and data wrangling. Engineered 3 new columns: Mood, Vocals, and Instrumentation.
- Loaded the dataset into Claude AI to perform LLM-assisted classification across the new metadata columns. Manually validated and sense-checked the tags.
- Re-imported the CSV into Python and built a multi-parameter filtering script to query and parse tracks based on arrayed mood lists and precise duration time windows.
- Prompted Claude AI to generate 3 sync briefs to serve as test parameters:

Brief 1: Coming-of-age film

Indie drama. Our two leads drive home after graduation, windows down, unsure what comes next. We want something warm and a little wistful that builds quietly, not sentimental. Vocal is fine. Looking for roughly 3:30–4:30 of usable music.

Brief 2: Club scene

TV drama, a packed club. Our lead spots her ex across the dance floor, laughing with someone new, and the night turns. We need something driving and slightly dark that works under dialogue, with a simmering, envious edge, and that can come up when the camera hits the dance floor and she watches them together. 1 to 2 minutes.

Brief 3: Really sad scene

Feature film, the final scene of a breakup. One character packs up the flat alone. We want something stripped emotional, intimate and raw. Sparse instrumentation, nothing that feels manipulative. Under 3 minutes.

- Programmed these query constraints into the Python script, which executed the search logic and returned a ranked set of 5 candidate tracks per brief.

Findings: Qualitative evaluation of the 5 output tracks per brief confirmed a high precision rate, with most recommendations strongly aligning with the target emotional, vocal, and duration parameters. I was happy with most of the tracks the algorithm returned.